The cost of a parametric Gröbner basis over a function field

Haniel Ulises

5 September 2026

Abstract

The inverse kinematics pipeline in `varietas` computes a Gröbner basis over $\mathbb{Q}(\boldsymbol{p})$, adjoining the target pose to the coefficient field so that one basis answers every pose. The computation completes in milliseconds with two adjoined parameters and has never been observed to complete with three. Earlier profiling attributed 86 % of the running time to `polynomial_gcd`, which raised the question of whether a modular gcd would remove the limit. This report answers that question by repeating the computation over $\mathbb{F}_p(\boldsymbol{p})$, where coefficient arithmetic costs one machine multiplication. The three-parameter problem does not complete over $\mathbb{F}_p$ either. A separate measurement of the gcd in isolation shows that the choice of field accounts for a bounded factor of about 13, while the cost grows as roughly the fourth power of the number of terms in the operands. The limit is therefore attributable to the growth of the parameter polynomials rather than to the size of their coefficients, and a modular gcd built on the existing subresultant sequence would not remove it.

1 Objective

The pipeline poses inverse kinematics with the target adjoined to the coefficient field. For an arm with N actuated joints and P adjoined pose coordinates, the ring is $\mathbb{Q}(\boldsymbol{p})[t_1, \ldots, t_N]$ with $\boldsymbol{p} = (p_1, \ldots, p_P)$, and a finite solution set requires $P = N$. The measured behaviour is

- $P = N = 2$: complete in 5 ms to 25 ms;
- $P = N = 3$: no result observed after 900 s.

Coefficients are elements of $\mathbb{Q}(\boldsymbol{p})$, represented as a quotient of two polynomials in $\boldsymbol{p}$ over $\mathbb{Q}$ and reduced to lowest terms after every arithmetic operation. The reduction requires a polynomial gcd, computed by a subresultant remainder sequence. Sampling an instrumented three-parameter run attributed approximately 86 % of the elapsed time to that gcd.

Two explanations are consistent with that measurement. Under the first, the cost lies in arithmetic on rational coefficients, whose numerators and denominators grow during the remainder sequence; a modular gcd would then remove most of it. Under the second, the cost lies in the number of terms in the parameter polynomials, which no change of coefficient representation affects. The objective of this experiment is to distinguish the two before committing to an implementation.

2 Method

2.1 Change of field

The discriminating measurement replaces $\mathbb{Q}$ by the prime field $\mathbb{F}_p$ with $p = 2^{31} - 1$, leaving the algorithm, the monomial order, the ideal, the number of parameters and the sequence of operations unchanged. In $\mathbb{F}_p$ an addition or multiplication is one machine-word operation followed by a remainder, and coefficients do not grow. Any cost that survives the substitution is independent of coefficient size.

The field $\mathbb{F}_p(\boldsymbol{p})$ is implemented in `doc/experiments/modular_field.hpp` as a quotient of two polynomials over $\mathbb{F}_p$, normalised in the same three steps used for $\mathbb{Q}(\boldsymbol{p})$: division by the monomial content, division by the polynomial gcd, and scaling to a monic denominator. The remainder of the library is templated on the coefficient type and is used without modification.

2.2 Systems measured

Three systems are posed, each as a position problem with one adjoined coordinate per joint.

Planar 2R. Two joints turning about z, unit links, adjoining (x, y). Two solutions.

Reduced 2R. The shoulder and elbow of an anthropomorphic arm, obtained by sweeping its base joint out, adjoining a radius and a height. Two solutions.

Anthropomorphic 3R. A base yawing about z and two joints pitching about y, unit links, adjoining (x, y, z). Four solutions, established independently by solving the same system over $\mathbb{Q}$ at a fixed pose.

The first two serve as controls. If the change of field alters the algebraic structure of the answer, the prime is unlucky for these inputs and the comparison is not valid.

2.3 Isolated gcd

A second program, `doc/experiments/gcd_cost.cpp`, measures a single gcd outside the pipeline. Two dense trivariate polynomials of equal degree are formed with a planted common factor, and the subresultant gcd is timed over both fields. Varying the degree separates the contribution of coefficient size from the contribution of operand size.

2.4 Environment

Measurements were taken on an Intel Core i7-10870H at 2.20 GHz, with g++ 11.4.0 at `-O2`, GMP for rational arithmetic. Times are single runs; the quantities of interest differ by orders of magnitude and are not sensitive to run-to-run variation.

3 Results

3.1 Controls

Table 1 reports the two-parameter systems. The basis size, the dimension of the quotient algebra and the largest polynomial in the basis agree exactly between the two fields, so the prime is not unlucky for these inputs and the substitution preserves the computation. The change of field reduces the Gröbner phase by a factor of about four.

Table 1: Two-parameter systems. Times in seconds. Structure of the answer is identical in both fields.

System	Field	Forward map	Gröbner	Basis	$\dim_k A$	Max terms
Planar 2R	$\mathbb{F}_p(x, y)$	0.000	0.006	2	2	3
Planar 2R	$\mathbb{Q}(x, y)$	0.001	0.023	2	2	3
Reduced 2R	$\mathbb{F}_p(r, z)$	0.000	0.006	2	2	3
Reduced 2R	$\mathbb{Q}(r, z)$	0.001	0.023	2	2	3

3.2 Three parameters

The anthropomorphic 3R system with (x, y, z) adjoined produced no result over $\mathbb{F}_p(x, y, z)$ within 400 s, nor over $\mathbb{Q}(x, y, z)$ within 900 s. Construction of the rational forward map and of the residuals completed in under 0.05 s in both fields, so the whole of the elapsed time is spent in the Gröbner computation.

Instrumenting the pair queue shows where. Over $\mathbb{F}_p(x, y, z)$ the run enters the main loop with four basis elements, six pending pairs and a largest basis polynomial of fourteen terms, and completes one pair. No second pair completed within the following 300 s. The run is therefore not accumulating a large basis; it is inside a single reduction.

Instrumenting the normalisation shows what grows during that reduction. The largest parameter polynomial appearing as a numerator or denominator reaches 460 terms after 93 s and 498 terms after 129 s, with intervals of tens of seconds between successive normalisations. Since coefficients in $\mathbb{F}_p$ do not grow, these intervals measure the cost of a gcd on trivariate operands of that size.

3.3 Isolated gcd

Table 2 reports the subresultant gcd on dense trivariate operands.

Table 2: One subresultant gcd on dense trivariate polynomials with a planted common factor.

Degree	Terms per operand	$\mathbb{F}_p$ (s)	$\mathbb{Q}$ (s)	Ratio
4	35	0.001	0.004	4.0
6	84	0.008	0.096	12.0
8	165	0.109	1.394	12.8
10	286	1.049	14.216	13.6

Two quantities are visible. The ratio between the fields rises and settles near 13, which bounds what any change of coefficient representation can recover. The cost over $\mathbb{F}_p$ grows steeply in the number of terms: the successive ratios of running time to the corresponding ratios of operand size give exponents of 2.4, 3.9 and 4.1, so the cost scales approximately as the fourth power of the operand size. Extrapolating the last measurement to the 498 terms observed in the three-parameter run gives approximately 9 s per gcd over $\mathbb{F}_p$ and 120 s over $\mathbb{Q}$, consistent with the intervals between normalisations reported above.

4 Analysis

The controls establish that the substitution of $\mathbb{F}_p$ for $\mathbb{Q}$ preserves the computation. The three-parameter system then fails to complete in a setting where coefficient arithmetic is free, which excludes the first of the two explanations under test: coefficient size is not what prevents the run from finishing.

The isolated measurement locates the remaining cost. The subresultant remainder sequence performs a number of polynomial multiplications and exact divisions that grows with the degree of the operands in the main variable, and each of those operations is itself performed on multivariate polynomials whose size grows with the recursion into the remaining variables. The observed fourth-power scaling is a property of that structure and is present over both fields.

The consequence for a modular gcd depends on which construction is meant. Computing the same subresultant sequence over several primes and reconstructing by the Chinese remainder theorem addresses coefficient growth alone, and Table 2 bounds the resulting improvement at about a factor of 13. Applied to a run that has not completed in 900 s, that factor does not produce a usable result. A gcd built on evaluation and interpolation, in the manner of Brown or Zippel, replaces the remainder sequence itself: the multivariate problem is reduced to univariate gcds at sampled points and the answer is interpolated, which changes the exponent rather than the constant. That construction addresses the cost identified here, and a multi-prime variant of the existing sequence does not.

Two qualifications apply. First, the extrapolation to 498 terms spans a factor of 1.7 beyond the largest measured operand and is used only to confirm consistency with the observed intervals. Second, the experiment establishes that a faster gcd is necessary for the three-parameter problem, and does not establish that it is sufficient. The parameter polynomials reach several hundred terms during a single reduction, and the number of reductions the run requires is unknown, since it has never completed. A gcd whose cost grew linearly rather than quartically would remove the present obstruction without guaranteeing that the run terminates in useful time.

5 Conclusions

1. The three-parameter parametric solve does not complete over $\mathbb{F}_p(x, y, z)$, so its cost is not attributable to arithmetic on rational coefficients.
2. The subresultant gcd over a fixed prime field costs approximately the fourth power of the number of terms in its operands, and the parameter polynomials reach 498 terms within one reduction of the three-parameter run.
3. A modular gcd formed by computing the existing remainder sequence over several primes is bounded by a factor of about 13 and would not make the three-parameter problem tractable.
4. A gcd based on evaluation and interpolation addresses the measured cost. Whether it is sufficient is not established by this experiment and would itself require measurement.
5. The recommendation is to treat $P = 2$ as the working limit and to obtain three-joint arms by the decoupling already implemented, which reduces such an arm to a two-parameter problem solved in 25 ms.

6 Reproduction

From the repository root:

```
g++ -std=c++17 -O2 doc/experiments/field_cost.cpp -o field_cost \
  -Ivarietas_core/include -Ivarietas_codegen/include \
  -Ivarietas_kinematics/include -Idoc/experiments \
  -I/usr/include/eigen3 -lgmpxx -lgmp

./field_cost planar      # two-parameter control, both fields
./field_cost reduced     # two-parameter control, both fields
./field_cost spatial-fp  # three parameters over F_p; does not terminate
./field_cost spatial-q   # three parameters over Q;   does not terminate

g++ -std=c++17 -O2 doc/experiments/gcd_cost.cpp -o gcd_cost \
  -Ivarietas_core/include -Ivarietas_codegen/include \
  -Idoc/experiments -I/usr/include/eigen3 -lgmpxx -lgmp

./gcd_cost
```

Adding `-DVARIETAS_EXPERIMENT_TRACE` to the first build reports the largest parameter polynomial seen during normalisation. The pair-queue figures were obtained with a temporary instrumented copy of `varietas_core/include/varietas/core/ideal/buchberger.hpp` printing the basis size, the pending count and the largest basis polynomial on each pass through the main loop.